

THEMBISILE MAKHUBU

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EDUCATION

BSC COMPUTER SCIENCE

Jan '22 - Present

Stellenbosch University | Stellenbosch

NATIONAL SENIOR CERTIFICATE

Jan '17 - Dec '21

Falesizwe Secondary School | Frankfort

EXPERIENCE

Software Engineering Job Simulation on Forage

Sep '24

Standard Bank

- Established a simulation project to enhance Standard Bank's mobile app, **doubling efficiency** through Java Web Token integration
- Executed the construction of a Spring Boot application with **JSON web tokens for enhanced authentication security measures** featuring token-based authentication and user access control management
- Assembled a Python Lambda function to **encode faces within images**, enhancing mobile authentication efficiency
- Leveraged SQL queries to extract data from Teradata, resulting in an **efficiency boost**
- Prepared a Django application to return a personalised **view of a user's data**

Class Representative

Mar '24 - Jun '24

Logistics Department

- Participated in organizational skills workshops during the Class Representative Training functions hosted by the Economic and Management Sciences Student Council (EBSK) to elevate leadership skills and hone communication abilities, resulting in **improved team productivity and enhanced decision-making processes**
- Advocated for the representation of academic interests of **class peers** to key bodies
- Facilitated cross-functional collaboration using Microsoft Teams, improving communication and **increasing efficiency** across student and academic bodies
- Ensured student perspectives were considered through obtaining and **relaying feedback** to the student and lecturers respectively

Back-End Engineering Job Simulation on Forage

May '23

Lyft

- Implemented unit testing and adopted test-driven development practices to ensure code reliability
- Coordinated comprehensive **code refactoring** projects, introducing modularization and adherence to coding standards
- Gained **hands-on experience** in Software Architecture, designing scalable and maintainable systems
- Revamped RESTful APIs for **performance and scalability** while employing Git version control for streamlined feature development

Tutoring Mathematics and Physics

Jan '23 - Feb '23

Falesizwe Secondary School

- Engineered tailored learning strategies to facilitate student comprehension of complex subjects and foster **advanced problem-solving skills** with a focus on interactive educational techniques
- Mentored students to excel in exam preparation, facilitating **top scores in test simulations**, quiz practice, and mock exams using tailored strategies
- Fostered a positive, inclusive learning environment by **adapting teaching strategies** to diverse learning styles, boosting engagement and student performance.

SKILLS

- **TECHNICAL:** Spring Boot, Django, AWS Lambda, RESTful APIs, SQL, Unit Testing, Test-Driven, Development, Software Architecture, Git, JSON Web Tokens, React, JavaScript, Java, Python, SQL, HTML/CSS, C, and Assembly
- **INTERPERSONAL:** Collaboration, Communication, Problem-Solving, Leadership, Adaptability, Time Management, Attention to detail

ACHIEVEMENTS

- **3rd Place – Standard Bank Group University Hackathon (Recess 2025)**
Collaborated in a team to develop, deploy, and present an **AI-powered wellness solution** tailored for Standard Bank customers — all within 48 hours. The platform enabled users to chat in their home language with others facing similar challenges, and access personalized recommendations from an AI coach that integrated Standard Bank's value-added services
- **Woman Regional Master (WRM), CHESSA – 2025**
Recognised by the Chess South Africa federation for competitive excellence in regional-level tournaments. Demonstrates advanced strategic thinking, focus, and sustained high performance.
- **Dell Young Leaders Scholarship Award (2022)**
Selected for the prestigious scholarship in recognition of resilience, ambition, and leadership potential. The award supports high-potential university students committed to driving change in their communities and excelling in their careers.
- **Top Performance – Computer Networking Projects (University Module)**
Developed multiple networked systems across four major projects, earning an overall average of **88.44%**:
 - Client-Server Chat App** – Designed and implemented a robust chat system; scored **96.25%**
 - RBUDP vs TCP File Transfer** – Built and analyzed a custom protocol; scored **84.71%**
 - VoIP System** – Created a multi-client voice communication app; scored **78.67%**
 - P2P File Sharing App** – Engineered a uTorrent-style P2P application; scored **93.75%**